

Owen Farmer

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Education

College of Engineering - University of California, Berkeley

December 2021

B.S. Mechanical Engineering

GPA: 3.4/4.0

Coursework: Manufacturing and Tolerancing (Reader), Rapid Prototyping, Design for Manufacturing, Engineering Drawings, Human Inspired Design/Factors, Manufacturing Improvement, Mechanical Validation, Statics, Fluids, Dynamics, Heat Transfer

Work Experience

Dolby Laboratories, Mechanical Engineer

September 2024 - Current

- Design speaker parts (plastic, metal, wood, and softgoods) - using TA's, FEA, hand calcs, testing to validate designs
- Share drawings, review build data/inspection reports, and provide onsite support with local suppliers and factories in China
- Integrate electromechanical components into devices such as screens, antennas, PCBs, wiring, and speaker modules
- Work with Acoustic/Electronic Engineers to define acoustic/heat/grounding requirements, create test fixtures, and validate designs

Nokia, Associate Mechanical Hardware Design Engineer

January 2023 - July 2024

- Concept and design die cast/machined metal, sheet metal, plastic, FIP gasket, & PCB component parts in multiple telecom radios
- Work with local and overseas vendors to generate 2D/3D drawings, submit requests for quotes, review DFMs & FAI data
- Create fixtures, determine metrics, & perform tests for water ingress, wind-driven rain, vacuum seal, board strain, and thermal
- Perform FEA, topology optimization, tolerance analysis, & use GD&T to virtualize portions of the design process

Facebook, Mechanical Engineering Intern, Server Hardware

June 2021 - September 2021

- Designed new server tray supporting add-in cards, creating sheet metal structure/supports, plastic baffles, die cast latches/handles
- Performed tolerance analysis to determine spacing between parts on PCB and between key locating features at tray and rack level

Apple, Product Design Intern, Audio Products

February 2021 - June 2021

- Developed fixtures to hold plastic housing & cosmetic mesh in place in order to identify step/gap issues, created TA for assembly
- Designed and validated a headphone pressure sensing fixture that simulated forces exerted on the side of a user's head
- Simplified CAD model for FEA drop simulations, creating glue/solder bodies, representations for flexes, PCBs, and defeatured module representations

Tesla Motors, Quality Engineering Intern, Body in White

January 2020 - April 2020

- Recertified/tuned framing fixtures to solve quality issues, creating documentation to coordinate changes between production shifts
 - Redesigned metal datum pins in framing stations, machining prototypes to test altered geometry
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Projects

Unicycle Frames and Accessories, Fabricator + Welder

December 2021 - Current

- Started my own side business of creating high-end custom mountain/street unicycles and unicycle related accessories
- Design, CAD, perform FEA analysis, machine, notch, TIG weld, and test each frame per customers' needs/requests, making them compatible with specific unicycle/bike parts that I recommend based on riding style and preference
- Create clamping/aligning fixtures for frames and parts during machining, notching, and welding processes

Human Powered Vehicle Team Lead

September 2017 - January 2021

- Led a team of 15 to design, manufacture, prototype, and test a human powered vehicle capable of travel at 60 mph
- Cut, bent, and welded aluminum/steel tubing, created CAD models/gcode and CNC'd various parts on the bike, created carbon layups for composite parts, assembled/sourced off the shelf bike components to use for our vehicle

Internally Geared Automatic Bike Hub, College Senior Project

August 2021 - December 2021

- Designed a bike hub that switches from an internal 1:1 drive and 4:3 internally geared drive based on the pedaling cadence
 - Machined all parts myself, including waterjet gears, lathed housings/axles/bushings, and milled drivetrain components
 - Created CAD models for all parts and 2D technical drawings for reference while machining parts
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Skills and Interests

Software: Creo/Windchill (passed Associate Certification Exam 2023), SolidWorks, NX/Teamcenter, ANSYS, Onshape, MS Office

Manufacturing Skills: Machining, TIG Welding, 3D Printing, Injection Molding, Sheet Metal, Die Casting, GD&T/2D Drawings

Languages: English (native), Mandarin (conversational speaking/listening, basic reading/writing)

Interests: Biking, Unicycling (World Record Holder, 3x North American Champion), Snowboarding, Skiing, Hiking